

- 8 (a) By reference to energy transfers, distinguish between electromotive force (e.m.f.) and potential difference (p.d.).

p.d. refers to the electrical energy per unit charge converted to other forms of energy [B1] whereas e.m.f. refers to the electrical energy per unit charge converted from other forms of energy. [B1]

.....[2]

- (b) A circuit is set up as shown in Fig. 8.1.

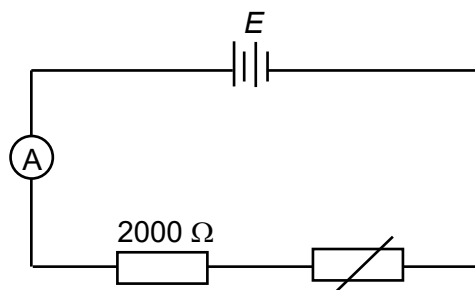


Fig. 8.1

The source of negligible internal resistance is found to provide 2.4×10^5 J of electrical energy to the 2000Ω resistor and thermistor when a charge of 2.2×10^4 C passes through the ammeter. At room temperature, the thermistor has a resistance of 1800Ω .

- (i) Sketch on Fig. 8.2 the variation with temperature θ of resistance R in a thermistor.

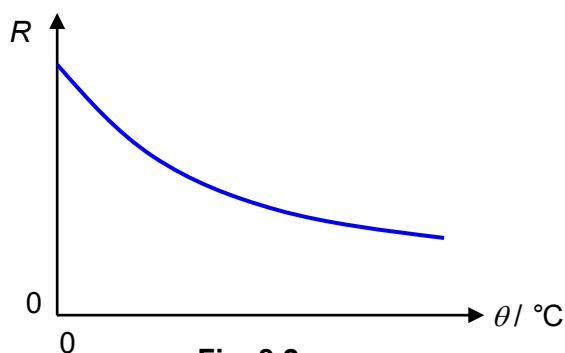


Fig. 8.2

Correctly drawn curve (straight lines not accepted)
cuts R-axis at 0°C .

[1]

(ii) For the thermistor at room temperature,

1. show that the e.m.f. of the source is 11 V.

$$E = \frac{W}{Q} = \frac{2.4 \times 10^5}{2.2 \times 10^4} \quad [\text{M1}]$$

$$= 11 \text{ V} \quad [\text{A0}]$$

[1]

2. determine the time taken for the charge of $2.2 \times 10^4 \text{ C}$ to pass through the ammeter.

$$Q = It = \left(\frac{E}{R}\right)t$$

$$t = \frac{QR}{E} = \frac{(2.2 \times 10^4)(2000 + 1800)}{11} \quad [\text{C1}]$$

$$= 7.6 \times 10^6 \text{ s} \quad [\text{A1}]$$

time = s [2]

3. determine the ratio

$\frac{\text{power dissipated in thermistor}}{\text{total power supplied by the cell}}$

$$\frac{P_{\text{therm}}}{P_{\text{total}}} = \frac{I^2 R_{\text{therm}}}{I^2 (R_{\text{therm}} + R_{2000})}$$

$$= \frac{1800}{1800 + 2000} \quad [\text{C1}]$$

$$= 0.47 \quad [\text{A1}]$$

ratio = [2]

- (c) A uniform resistance wire PQ of length 1.2 m is subsequently connected across the resistor and thermistor, as shown in Fig. 8.3. An ideal voltmeter is connected between point Y and a moveable contact M on the wire.

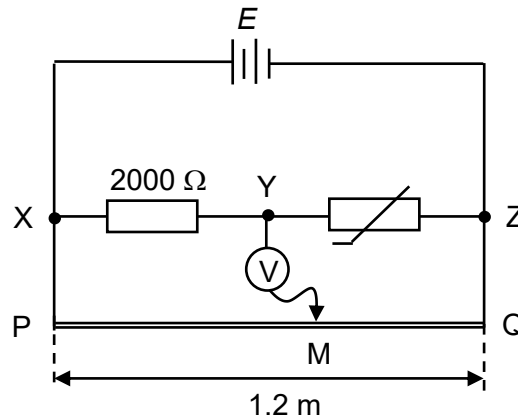


Fig. 8.3

- (i) At room temperature, the contact M is moved along PQ until the voltmeter shows zero reading.

Calculate the length of wire between M and Q

$$\begin{aligned}
 V_{MQ} &= V_{YZ} \text{ when voltmeter has zero reading} \\
 \frac{L}{1.2} &= \frac{1800}{2000 + 1800} \quad [\text{C1}] \\
 L &= 0.57 \text{ m} \quad [\text{A1}]
 \end{aligned}$$

length of wire = m [2]

- (ii) State and explain the effect, if any, on the length of the wire between M and Q for the voltmeter to remain at zero deflection if each of the following changes takes place independently.

1. The thermistor is warmed slightly.

With a rise in temperature, resistance of thermistor decreases. The ratio of p.d. across thermistor to the fixed resistor decreases, [M1] thus ratio of length MQ to PQ decreases too. Length MQ decreases. [A1]

.....[2]

2. A uniform wire of the same material but with a larger cross sectional area is used for PQ.

Since the total potential difference and total length remains the same, the potential gradient remains the same. [M1] Therefore length MQ remains the same. [A1]

.....[2]

- (d) The heating element of an electric heater is made of nichrome wire. Nichrome has a resistivity of $1.0 \times 10^{-6} \Omega \text{ m}$ at the operating temperature of the heater. The heater is rated at 240 V, 1200 W.

- (i) Determine the resistance of the nichrome wire when the heater is operating normally.

$$P = \frac{V^2}{R}$$

$$R = \frac{240^2}{1200} \quad [\text{C1}]$$

$$= 48 \Omega \quad [\text{A1}]$$

resistance = Ω [2]

- (ii) Calculate the length of nichrome wire of diameter 0.40 mm required for the heater.

$$R = \frac{\rho L}{A}$$

$$L = \frac{(48)(\pi(\frac{0.40 \times 10^{-3}}{2})^2)}{1.0 \times 10^{-6}} \quad [\text{C1}]$$

$$= 6.0 \text{ m} \quad [\text{A1}]$$

length of wire = m [2]

- (iii) The potential difference across the heater is then reduced to 180 V. Assuming the resistance of the nichrome wire remains constant, state and explain how this change affects the time taken to dissipate the same amount of thermal energy.

Since $P = \frac{V^2}{R}$, a decrease in potential difference will result in a decrease in power. [M1] Hence, the time taken to dissipate the same amount of heat would increase. [A1]

..... [2]

[Total: 20]

